

ThesisSphere

an Intelligent Thesis Management and Collaboration Platform

Software Requirements Specification (SRS)

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the design and development requirements for **ThesisSphere**, an Intelligent Thesis Management and Collaboration Platform developed using the **MERN Stack (MongoDB, Express.js, React.js, and [Node.js](#))**.

The system aims to simplify the thesis lifecycle by providing a centralized platform where students and supervisors can collaborate efficiently. It supports supervisor selection, topic management, research collaboration, progress tracking, paper reviews, AI-assisted recommendations, report generation, and virtual meetings.

1.2 Scope

The system provides a centralized platform for managing undergraduate and postgraduate thesis projects.

It supports:

- Student registration and thesis management
- Supervisor management
- Thesis topic publishing and selection
- Research collaboration
- Progress monitoring
- Online meetings
- AI-powered recommendations
- Report generation
- Paper review workflow
- Academic communication

The system serves four major user groups:

- Students
- Supervisors
- Department Administrators
- System Administrators

1.3 Definitions, Acronyms and Abbreviations

MERN

MongoDB, Express.js, React.js, Node.js

AI

Artificial Intelligence

JWT

JSON Web Token

REST API

Representational State Transfer Application Programming Interface

RBAC

Role-Based Access Control

1.4 References

- MongoDB Documentation
- Express.js Documentation
- React Documentation
- Node.js Documentation

1.5 Overview

This document specifies the functional requirements, non-functional requirements, architectural design, technology stack, and development methodology of ThesisSphere.

2. Overall Description

2.1 Product Perspective

ThesisSphere is a standalone web application that digitizes the thesis supervision process.

Instead of relying on manual paperwork, emails, and spreadsheets, the platform centralizes all thesis-related activities into a single collaborative environment.

The system integrates:

- Authentication
- AI Recommendation System
- Online Meeting
- Report Generator
- Messaging
- Notifications
- Progress Tracking

2.2 Product Features

No.	Feature	Description
1	Student Dashboard	Provides students with an overview of thesis progress, tasks, and upcoming activities.
2	Faculty Dashboard	Displays supervisees, pending reviews, meetings, and research activities.
3	Find Supervisor	Helps students search for supervisors based on expertise and research interests.
4	Discussion Forum	Enables academic discussions, questions, and knowledge sharing among users.
5	Thesis Groups	Allows students to view their group or browse other research groups.
6	Browse Topics	Lets students explore available thesis and research topics.
7	Meeting Scheduler	Schedules meetings between students and supervisors with reminders.
8	Messages	Supports direct communication between students and faculty members.
9	Deadline Calendar	Displays important submission dates, meetings, and academic deadlines.

10	Automated Report Generator	Generates progress reports and thesis documentation automatically.
11	Paper Review Manager	Manages paper submissions, reviews, revisions, and feedback.
12	Contribution Tracker	Tracks each group member's contributions throughout the project.
13	Notifications	Sends real-time alerts for meetings, deadlines, and important updates.
14	Student Profile	Displays a student's research interests, skills, and academic achievements.
15	Post Topics and Manage Applications	Allows supervisors to publish available thesis or research topics and students to apply for those topics and manage them.
16	Supervisor Profile	Displays supervisor expertise, research areas, and availability.
17	Thesis Group & Student Manager	Enables supervisors to organize thesis groups and manage assigned students.
18	AI-Based Recommendation Engine	Recommends supervisors, research topics, and papers based on user preferences.
19	Online Video Call Meeting	Provides integrated virtual meetings for thesis discussions and supervision.
20	Citation Generator	Automatically creates citations in multiple academic formats such as APA, IEEE, and MLA.

2.3 User Classes and Characteristics

Students

- Browse thesis topics
- Find supervisors
- Join thesis groups
- Submit reports
- Attend meetings
- Receive feedback

Supervisors

- Publish research topics
- Supervise students
- Review reports
- Schedule meetings
- Monitor progress

Department Administrator

- Manage departments
- Assign supervisors
- Monitor thesis progress
- Generate reports

System Administrator

- Manage users
- Maintain the system
- Configure permissions
- Monitor platform usage

2.4 Operating Environment

- Modern Web Browsers
- Desktop and Mobile Responsive
- MongoDB Database
- Node.js Server
- Cloud Deployment (AWS/Azure/Render)

2.5 Constraints

- Secure authentication
- Role-based permissions
- Stable internet connection
- AI recommendation requires sufficient training data
- Video meeting depends on WebRTC or third-party APIs

2.6 Assumptions and Dependencies

- Users have internet access.
- Supervisors regularly update available thesis topics.
- AI recommendations improve as more data becomes available.
- Email service is available for notifications.
- Video meeting service is operational.

3. System Requirements

3.1 Functional Requirements

Authentication & Authorization

FR-1

The system shall allow secure registration and login.

FR-2

The system shall support JWT-based authentication.

FR-3

The system shall implement role-based access control.

FR-4

The system shall provide separate dashboards for students, supervisors, administrators, and department coordinators.

Student Management

FR-5

Students shall browse thesis topics.

FR-6

Students shall request supervisors.

FR-7

Students shall create or join thesis groups.

FR-8

Students shall submit thesis reports.

FR-9

Students shall monitor progress.

Supervisor Management

FR-10

Supervisors shall publish thesis topics.

FR-11

Supervisors shall accept or reject supervision requests.

FR-12

Supervisors shall assign milestones.

FR-13

Supervisors shall review submitted documents.

FR-14

Supervisors shall provide feedback.

Collaboration

FR-15

The system shall provide direct messaging.

FR-16

The system shall support discussion forums.

FR-17

The system shall support online meetings.

FR-18

The system shall schedule meetings.

FR-19

The system shall send meeting reminders.

Thesis Management

FR-20

The system shall generate automated progress reports.

FR-21

The system shall track group contributions.

FR-22

The system shall manage thesis milestones.

FR-23

The system shall support version history for uploaded documents.

FR-24

The system shall maintain thesis archives.

AI Features

FR-25

The system shall recommend supervisors.

FR-26

The system shall recommend research topics.

FR-27

The system shall recommend research papers.

FR-28

The system shall generate citation references.

3.2 Non-Functional Requirements

Performance

NFR-1

Average page loading time shall be under 3 seconds.

NFR-2

Support at least 500 concurrent users.

Security

NFR-3

Passwords shall be encrypted.

NFR-4

All communications shall use HTTPS.

NFR-5

Role-based authorization shall be enforced.

NFR-6

Uploaded documents shall be securely stored.

Reliability

NFR-7

System uptime shall be at least 99.9%.

NFR-8

Automatic database backups shall occur daily.

Maintainability

NFR-9

The system shall follow modular architecture.

NFR-10

REST APIs shall be documented

Scalability

NFR-11

Support horizontal scaling.

NFR-12

Database shall support future expansion.

3.3 External Interface Requirements

User Interface

- Responsive React application
- Separate dashboards
- Interactive calendar
- Research analytics
- AI recommendation interface

Software Interfaces

- MongoDB
- Express API
- Node.js
- JWT Authentication
- Socket.IO
- WebRTC
- Google Calendar API
- Email Service API
- AI Recommendation API

Communication Interface

REST APIs

WebSocket (Real-time messaging)

WebRTC (Video Meeting)

4. Technology Stack & Architectural Overview

4.1 MERN Stack Components

MongoDB

Stores user information, thesis records, research topics, messages, and reports.

Express.js

Handles backend REST APIs.

React.js

Develops responsive user interfaces.

Node.js

Executes backend services.

5. Tentative Development Plan

Sprint 1

- Student Profile
- Faculty/Supervisor Profile
- Student Dashboard
- Faculty Dashboard
- Messaging

6. Conclusion

This Software Requirements Specification outlines the development of **ThesisSphere**, an Intelligent Thesis Management and Collaboration Platform designed to modernize academic thesis supervision. By integrating supervisor selection, thesis management, AI-powered recommendations, collaborative tools, automated reporting, and virtual communication within a unified MERN-based architecture, the platform enhances the efficiency and transparency of the thesis lifecycle. Following the proposed Agile development plan, ThesisSphere aims to provide a secure, scalable, and user-friendly solution that supports students, supervisors, and academic administrators throughout the research process.

Sprint 1 Screenshots:

Faculty Profile: Faculty Can access and edit his supervising profile from here.

The screenshot displays the ThesisSphere Supervisor Dashboard for a faculty member named Dr. Z. The dashboard is divided into a sidebar on the left and a main content area. The sidebar includes navigation links for Student, Faculty, Workload Dashboard, Post Topics, Meetings, Messages, Calendar, Paper Reviews, Notifications, and My Profile. The main content area features a header with the supervisor's name, title, and a button to edit the profile. Below the header are four summary cards: STUDENTS (3/5 supervising), RATING (4.9/5.0, 24 reviews), PUBLICATIONS (1, h-index: 32), and EXPERIENCE (12 yrs, Supervising since 2014). The BIOGRAPHY section states the supervisor is interested in deep learning and computer vision only. The RESEARCH AREAS section lists various fields: Cyber Security, Machine Learning, Deep Learning, Software Engineering, Natural Language Processing, Cloud Computing, Computer Vision, Data Mining, Internet of Things, and Blockchain. The CONTACT section provides email, address, website, and availability information.

ThesisSphere Supervisor Dashboard

SUPERVISOR PROFILE

DZ Dr. Z
Associate Professor · CSE · BRAC University

Accepting Students > Edit Profile

STUDENTS
3/5 supervising

RATING
4.9/5.0
24 reviews

PUBLICATIONS
1
h-index: 32

EXPERIENCE
12 yrs
Supervising since 2014

BIOGRAPHY
Interested in deep learning and computer vision only

RESEARCH AREAS

- Cyber Security
- Machine Learning
- Deep Learning
- Software Engineering
- Natural Language Processing
- Cloud Computing
- Computer Vision
- Data Mining
- Internet of Things
- Blockchain

CONTACT

- z@gmail.com
- 4G 10L, BRAC University
- carter-lab.mit.edu
- Fri 10-2pm
- Mode: Campus

This screenshot shows the same ThesisSphere Supervisor Dashboard, but with the 'Edit Faculty Profile' modal open. The modal allows the supervisor to update their designation, office room, consultation hours, availability mode, research topics, expertise keywords, publications, and profile bio. The background dashboard content is visible but slightly dimmed.

ThesisSphere Supervisor Dashboard

Edit Faculty Profile
Update topics, consultation hours, and availability mode.

Designation
Associate Professor

Office Room
4G 10L

Consultation Hours
Fri 10-2pm

Availability Mode
Campus Available

Topics
Select the main research topics you supervise.

- Machine Learning
- Deep Learning
- Natural Language Processing
- Computer Vision
- Data Mining
- Cyber Security
- Software Engineering
- Cloud Computing
- Internet of Things
- Blockchain

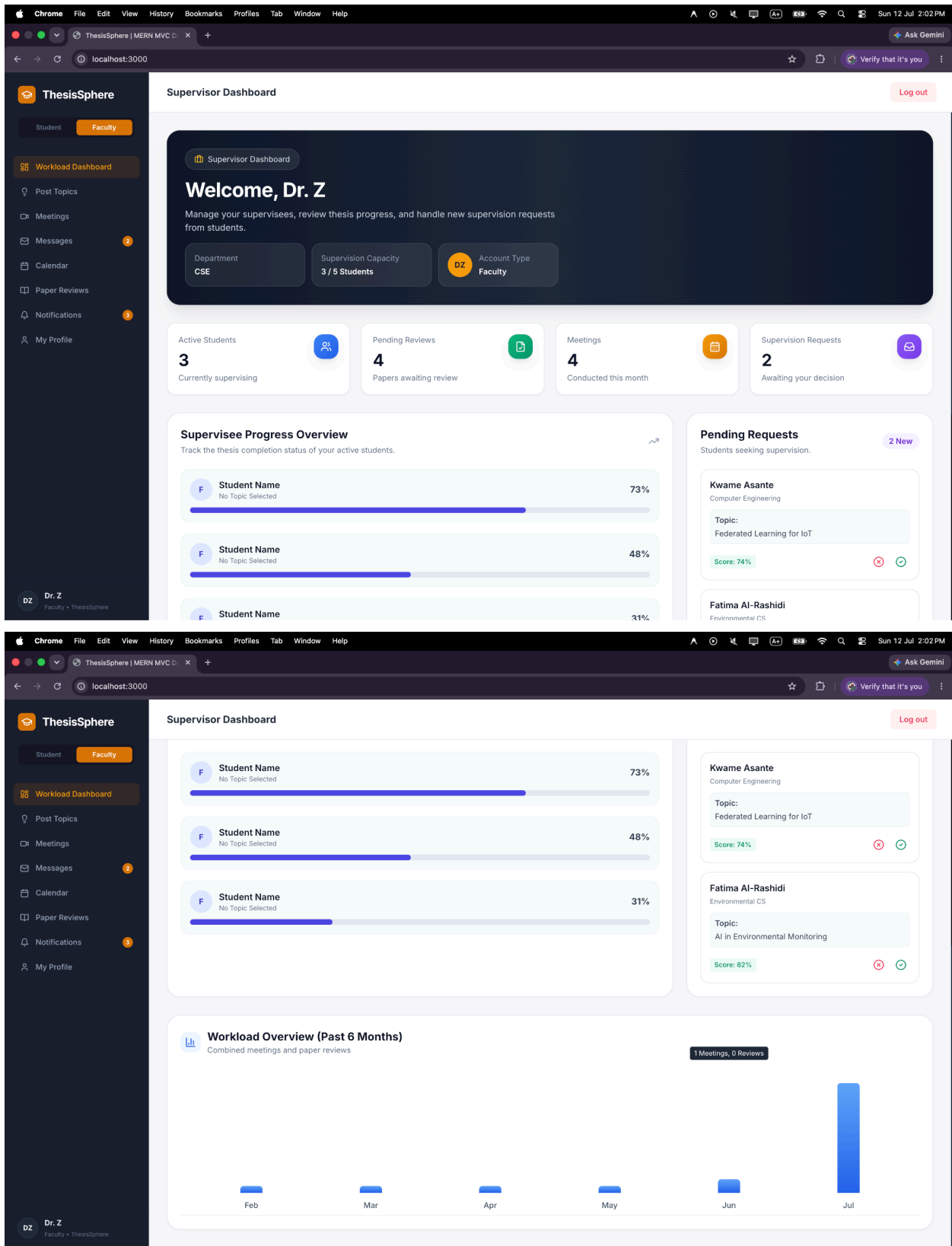
Expertise Keywords
Add comma-separated expertise keywords

Publications
Modified MASK R-CNN

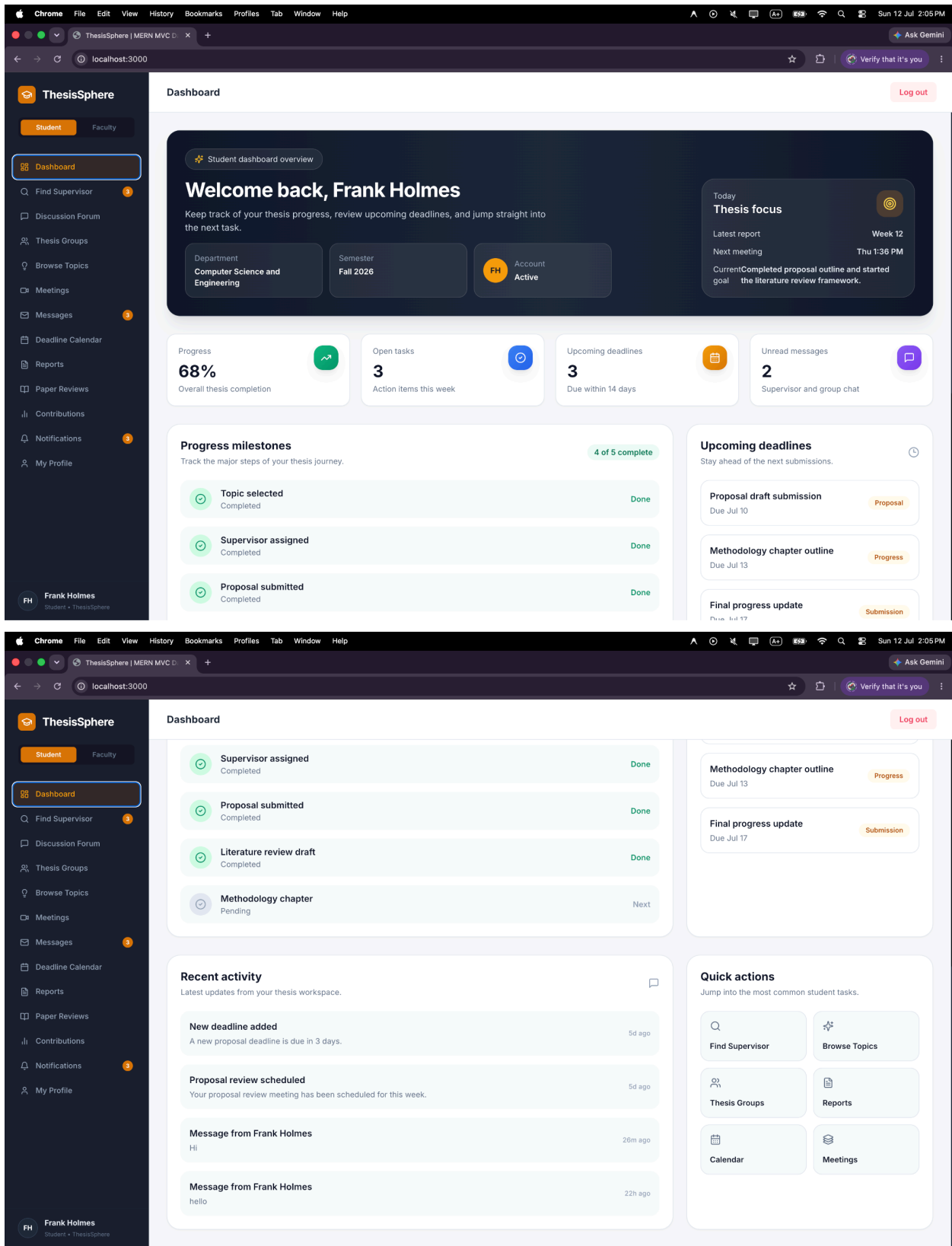
Profile Bio
Interested in deep learning and computer vision only

Cancel Save Changes

Faculty/Supervisor Dashboard: Faculty can track his work and review everything from here.



Students Dashboard: Students can do everything from this single dashboard and track his work.



Student Profile: Student can view and edit his profile from here.

The screenshot shows a web browser displaying the ThesisSphere application. The browser's address bar shows the URL `localhost:3000/profile`. The application has a dark sidebar on the left with the ThesisSphere logo and a navigation menu. The main content area is titled "Dashboard" and displays the profile of Frank Holmes, a student at BRAC University. The profile includes a blue circular avatar, a name, a department, and an "Edit Profile" button. Below the profile header are three sections: "Academic Information" (a table with Program, Semester, CGPA, and Publications), "Thesis Information" (Thesis Title and Supervisor), and "Research Interests" (a list of tags). At the bottom is a "Technical Skills" section with a list of skill tags. The sidebar menu includes options like Dashboard, Find Supervisor, Discussion Forum, Thesis Groups, Browse Topics, Meetings, Messages, Deadline Calendar, Reports, Paper Reviews, Contributions, Notifications, and My Profile (which is highlighted). The bottom of the sidebar shows the user's name, Frank Holmes, and their role as a student.

ThesisSphere

Student Faculty

Dashboard

Find Supervisor

Discussion Forum

Thesis Groups

Browse Topics

Meetings

Messages

Deadline Calendar

Reports

Paper Reviews

Contributions

Notifications

My Profile

Frank Holmes
Student • ThesisSphere

Dashboard

Frank Holmes
Computer Science and Engineering
BRAC University

Academic Information

Program	Bachelors
Semester	Fall 2026
CGPA	3.66
Publications	3

Thesis Information

Thesis Title
xai land

Supervisor
6a4cac7cc82195d44c38ba9d

Research Interests

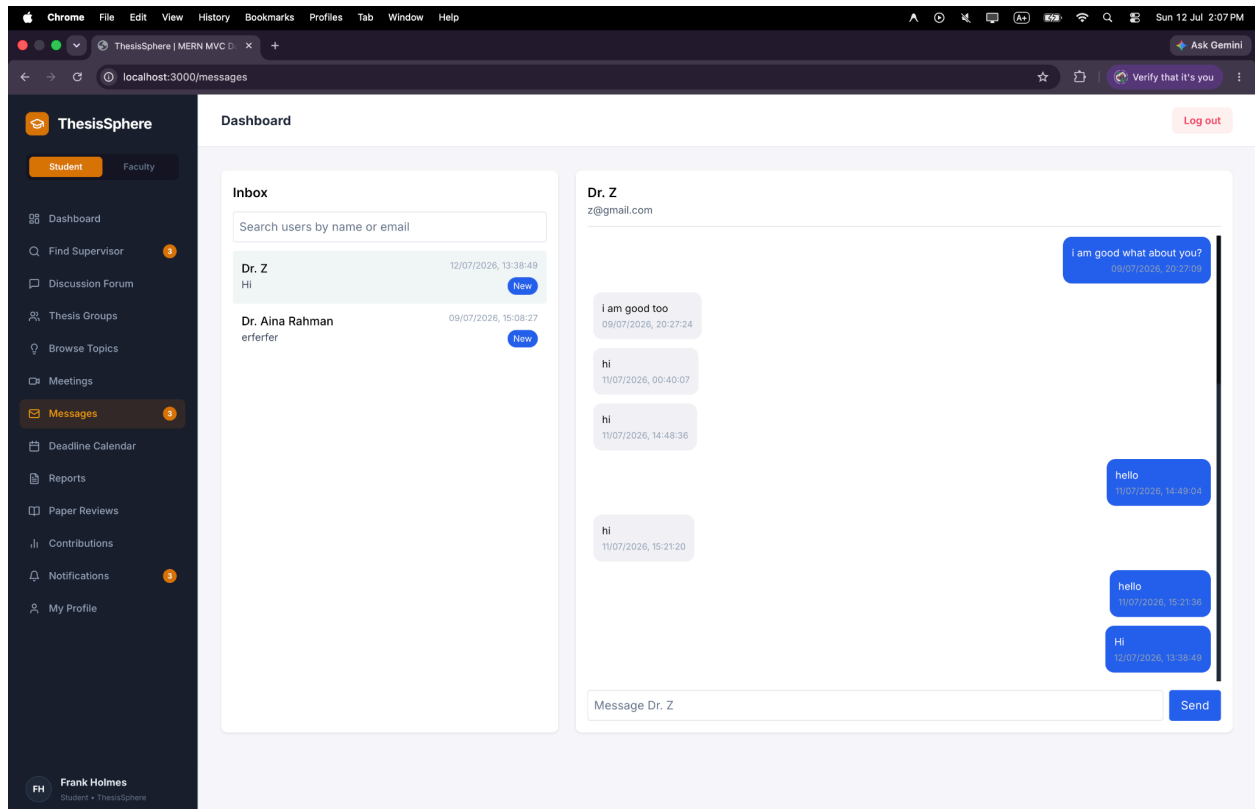
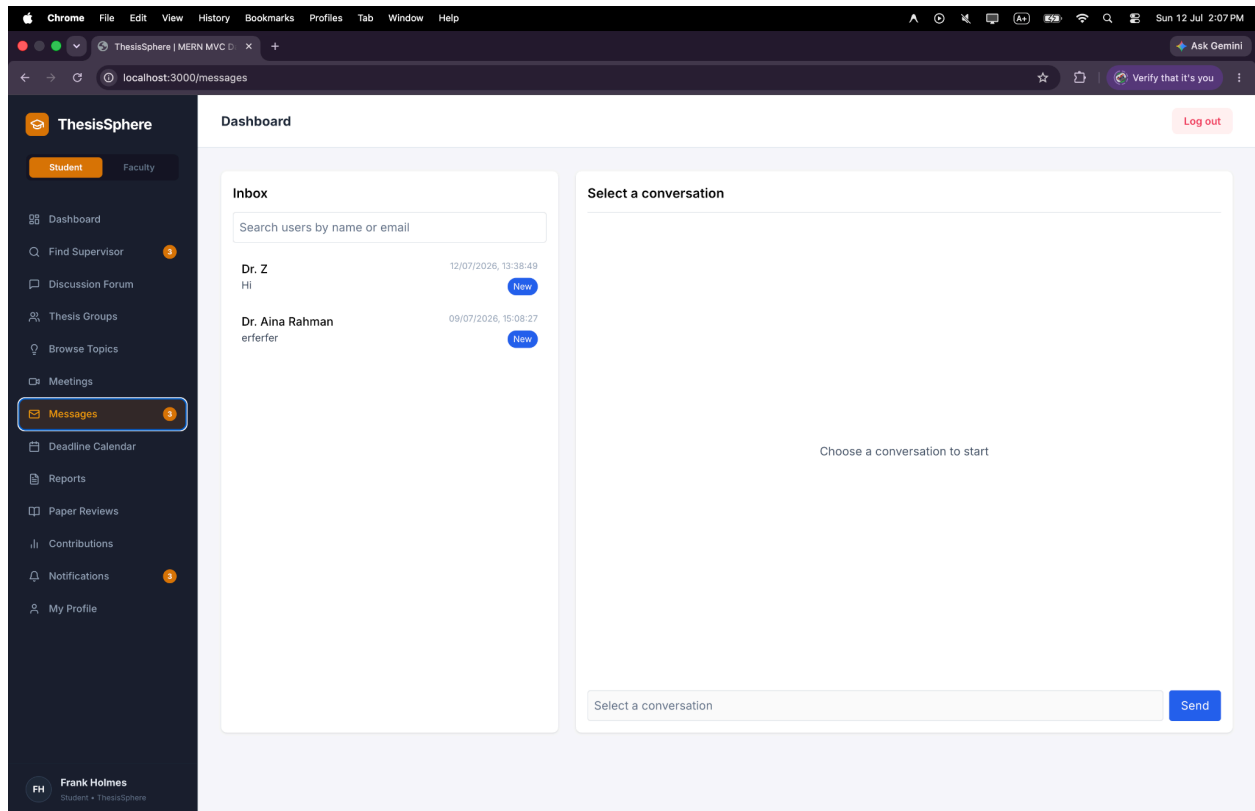
Machine Learning Data Mining Blockchain Cyber Security Internet of Things

Technical Skills

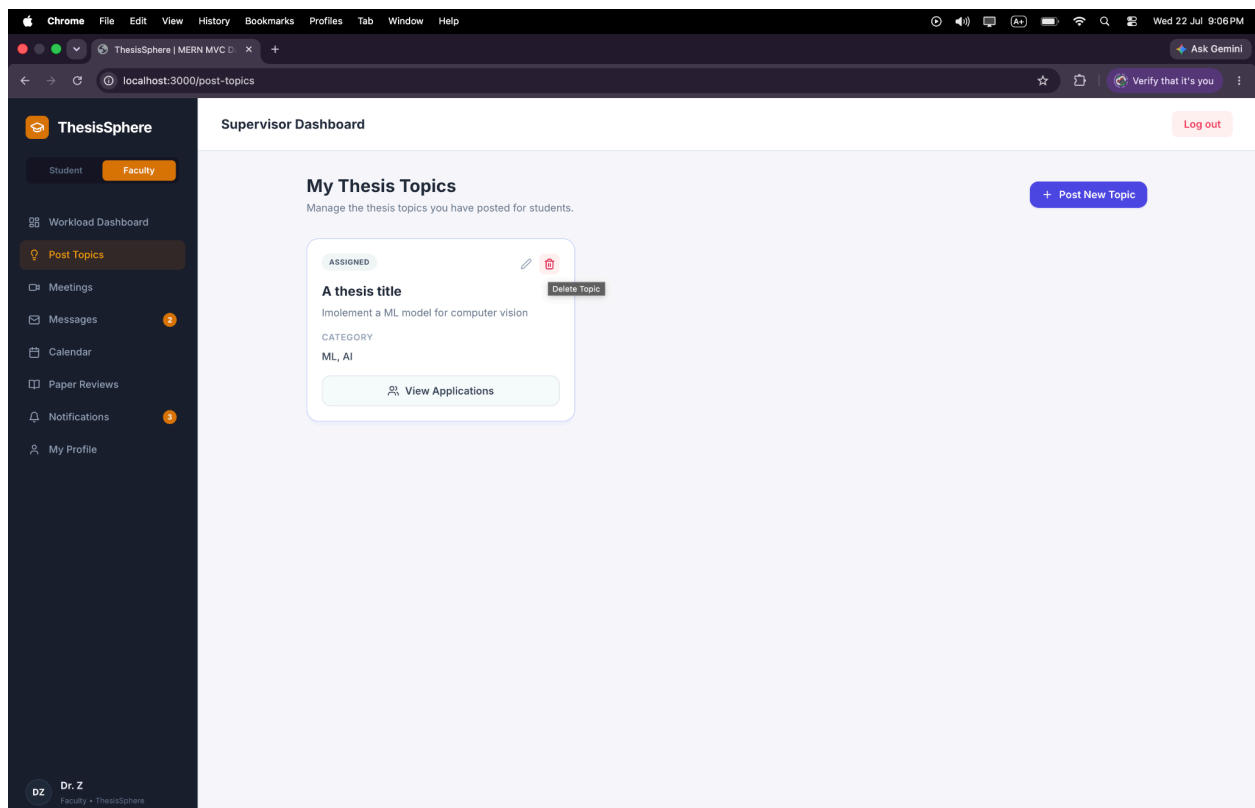
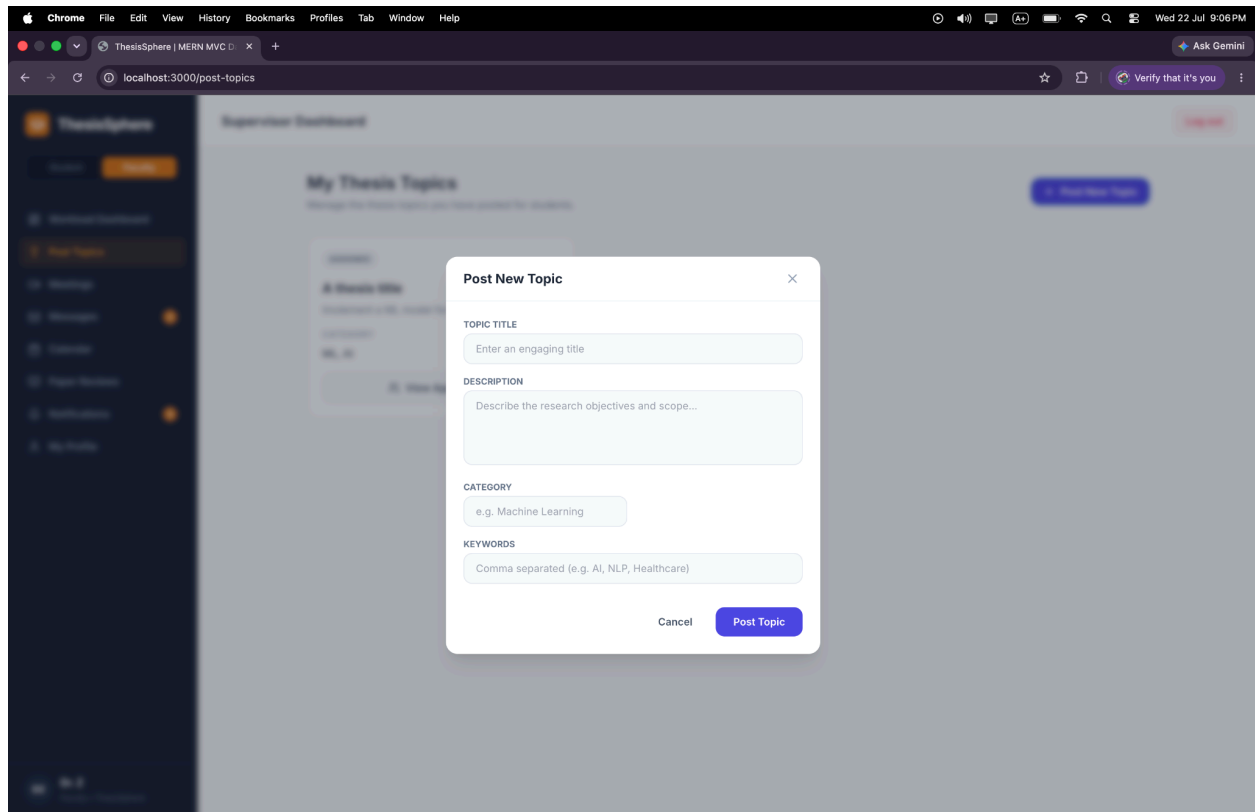
Research Data Analysis React MongoDB Git JavaScript C Python Java C++ Node.js TensorFlow

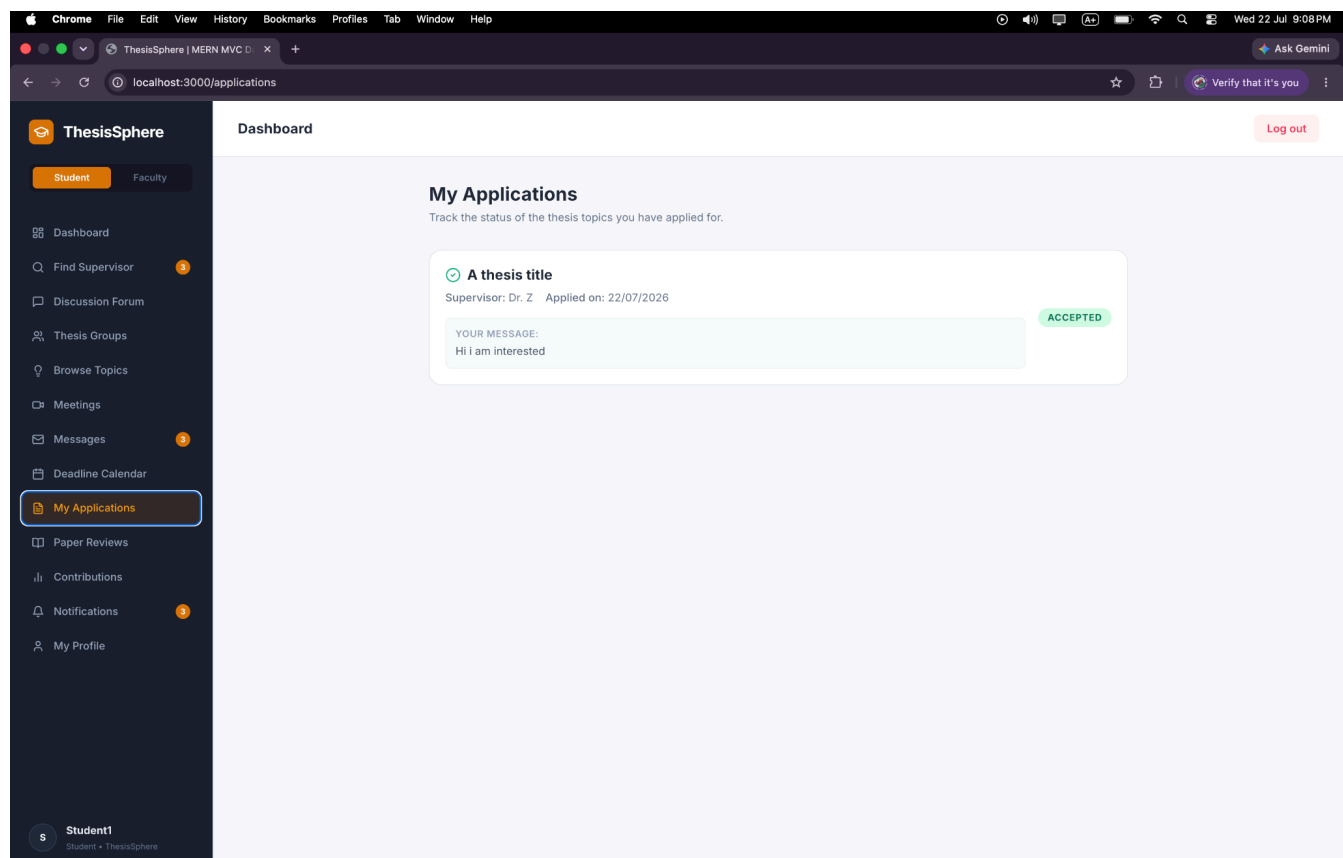
Log out

Messaging: Both the users can search and message each other.

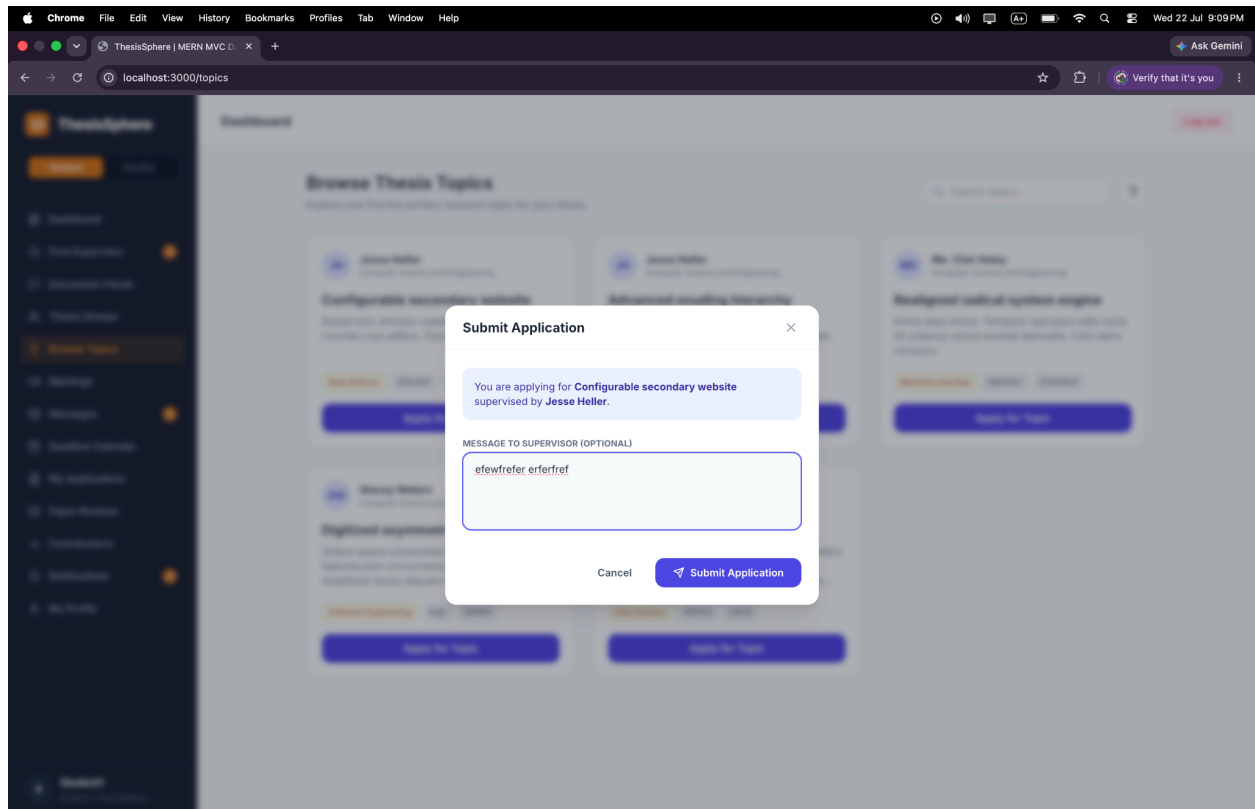
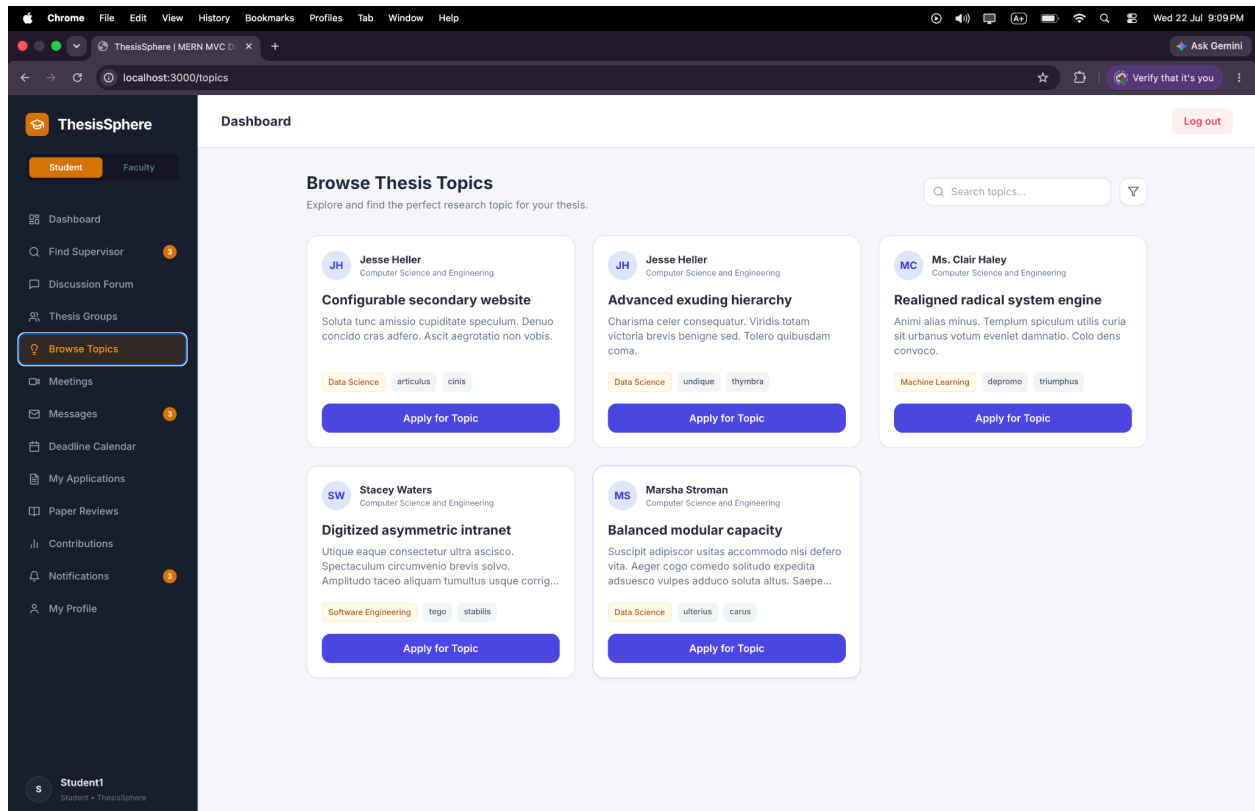


Post Topics and Manage Applications: Supervisor Can post, edit and delete topics from this portal and students can manage the application from their side.

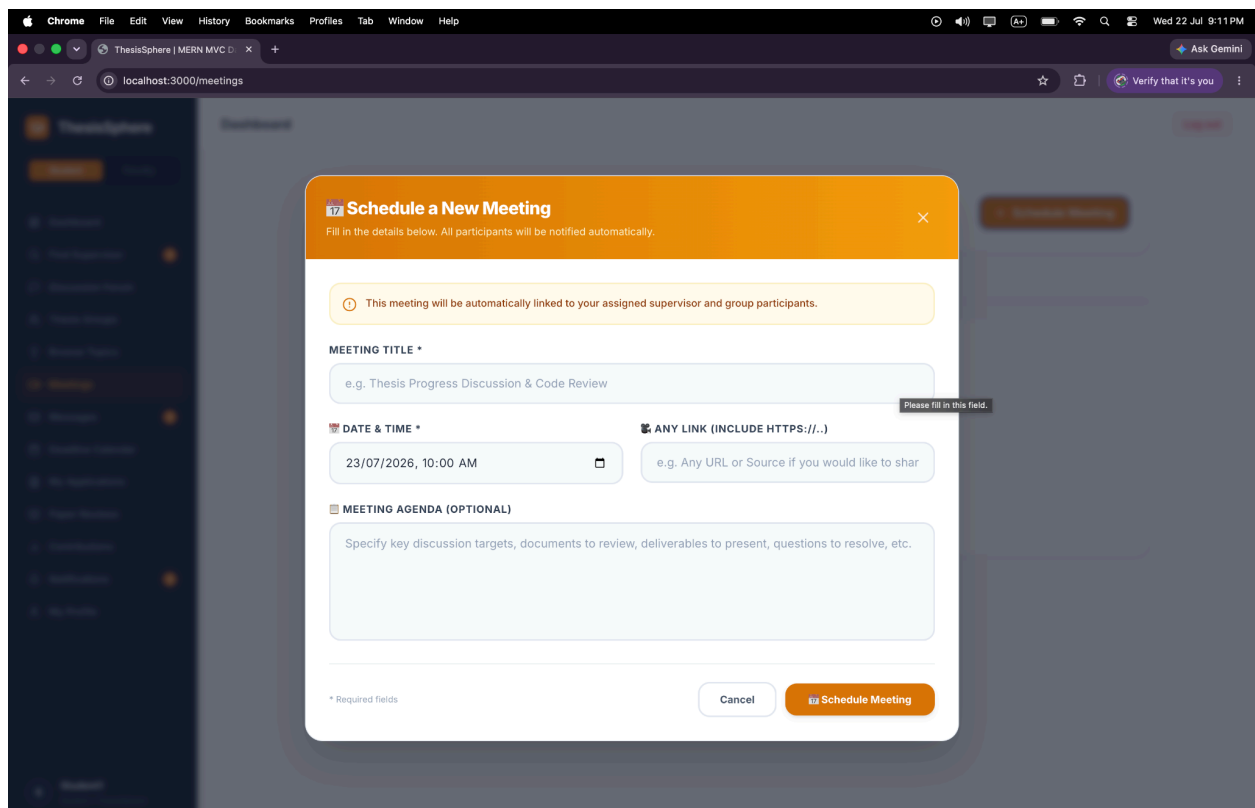
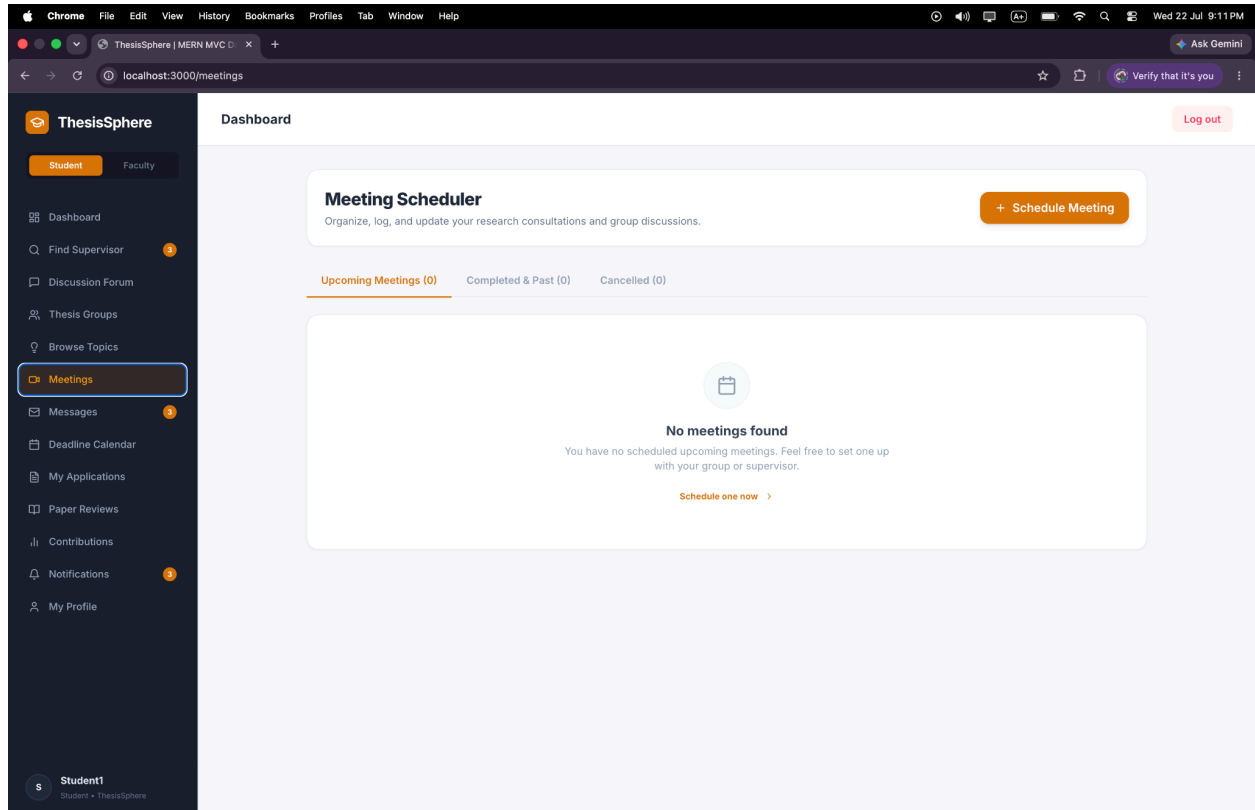


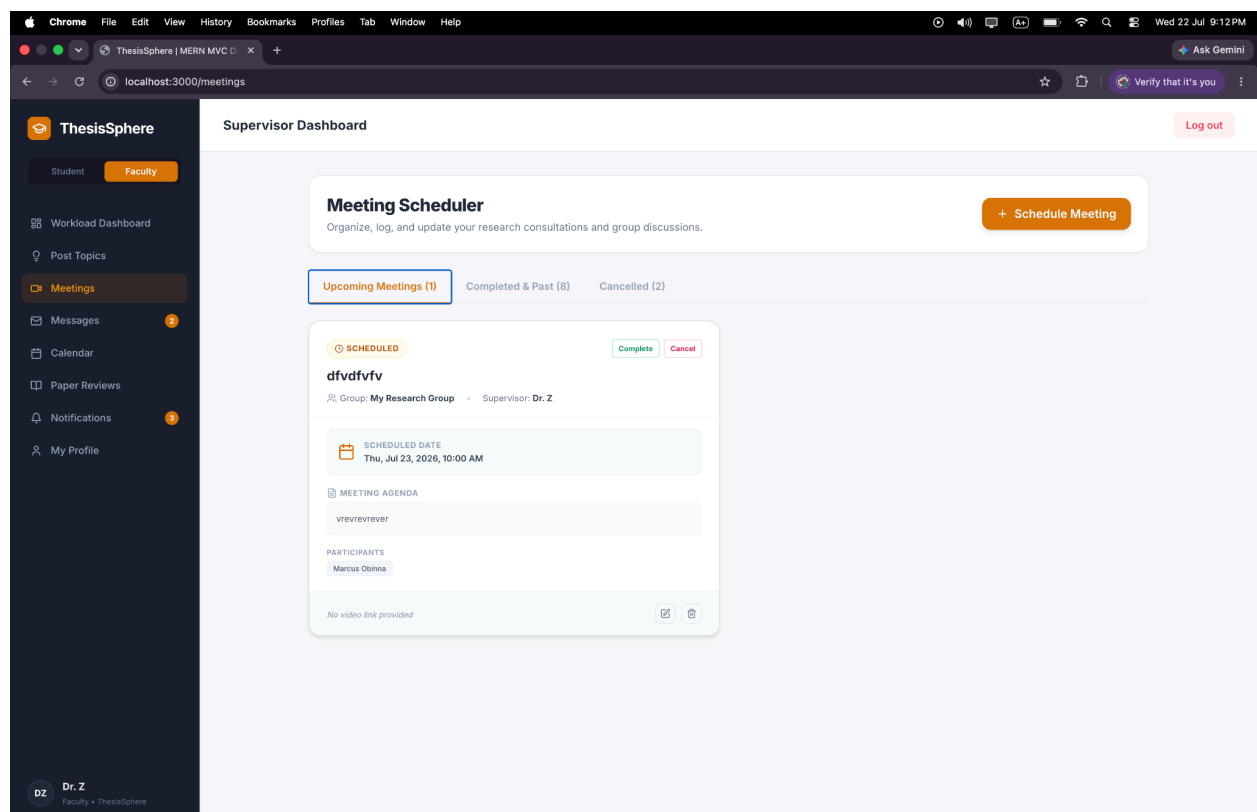


Browse Topics: Students can browse and apply for a specific topic from this portal.



Meeting Scheduler: Both the students and faculty members can schedule meetings with supervisor and students thesis group respectively using this portal.





ThesisSphere Supervisor Dashboard Log out

Meeting Scheduler
Organize, log, and update your research consultations and group discussions. + Schedule Meeting

Upcoming Meetings (1) **Completed & Past (8)** Cancelled (2)

COMPLETED

Project kick-off

Group: Deep Learning for Medical Image Analysis · Supervisor: Not assigned

SCHEDULED DATE
Fri, Jun 12, 2026, 10:46 AM

No video link provided

COMPLETED

Mid-term Review

Group: Deep Learning for Medical Image Analysis · Supervisor: Not assigned

SCHEDULED DATE
Sun, Jul 12, 2026, 10:46 AM

No video link provided

NEEDS REVIEW Complete Cancel

Chapter 3 Discussion

Group: Federated Learning in Healthcare Networks · Supervisor: Not assigned

SCHEDULED DATE
Sun, Jul 19, 2026, 10:46 AM

No video link provided

NEEDS REVIEW Complete Cancel

Literature Review Check

Group: Vision Transformers for Histopathology · Supervisor: Not assigned

SCHEDULED DATE
Sun, Jul 19, 2026, 10:46 AM

No video link provided

NEEDS REVIEW Complete Cancel

COMPLETED

Dr. Z
Faculty · ThesisSphere

ThesisSphere Supervisor Dashboard Log out

Meeting Scheduler
Organize, log, and update your research consultations and group discussions. + Schedule Meeting

Upcoming Meetings (1) Completed & Past (8) **Cancelled (2)**

CANCELLED

Thesis Process

Group: Federated Learning in Healthcare Networks · Supervisor: Not assigned

SCHEDULED DATE
Thu, Jul 23, 2026, 10:00 AM

MEETING AGENDA
N/A

PARTICIPANTS
Marcus Obinna

No video link provided

CANCELLED

Thesis Process

Group: Vision Transformers for Histopathology · Supervisor: Not assigned

SCHEDULED DATE
Tue, Jul 28, 2026, 12:02 PM

MEETING AGENDA
N/A

PARTICIPANTS
Ji-ho Park

No video link provided

Dr. Z
Faculty · ThesisSphere

Search Supervisor and AI assistance: Students can search supervisor from this portal and there will be an AI assistance feature for finding a good supervisor with

interest matching.

ThesisSphere

StudentFaculty

Dashboard

Find Supervisor3

Discussion Forum

Thesis Groups

Browse Topics

Meetings

Messages3

Deadline Calendar

My Applications

Paper Reviews

Contributions

Notifications3

My Profile

Student1
Student • ThesisSphere

Dashboard

Log out

Find Supervisor

Discover faculty members, explore their research interests, and find the perfect match for your thesis.

Search by name, department, or expertise...

Filters

AI-Recommended for You

Model: Local Algorithmic Math (Fallback)

Time: 636ms

Parsed Input: None

Top Supervisor Matches

J

Jesse Heller

Computer Science and Engineering • 0% Match

Machine LearningData Science

J

Jodie Boyer

Computer Science and Engineering • 0% Match

M

Ms. Clair Haley

Computer Science and Engineering • 0% Match

Top Topic Matches

Balanced modular capacity

By Marsha Stroman

0% Match

Digitized asymmetric intranet

By Stacey Waters

0% Match

Realigned radical system engine

By Ms. Clair Haley

0% Match

All Supervisors Directory

ThesisSphere

StudentFaculty

Dashboard

Find Supervisor3

Discussion Forum

Thesis Groups

Browse Topics

Meetings

Messages3

Deadline Calendar

My Applications

Paper Reviews

Contributions

Notifications3

My Profile

Student1
Student • ThesisSphere

Dashboard

Log out

All Supervisors Directory

Jesse Heller

International Configuration Developer

Dept: Computer Science and Engineering

Capacity: 0/6

EXPERTISE

taediumcanis

View Profile

Jodie Boyer

Global Markets Agent

Dept: Computer Science and Engineering

Capacity: 0/6

EXPERTISE

delegocornu

View Profile

Ms. Clair Haley

Investor Group Associate

Dept: Computer Science and Engineering

Capacity: 0/7

EXPERTISE

demorortantillus

View Profile

Marsha Stroman

Global Program Strategist

Dept: Computer Science and Engineering

Capacity: 0/7

EXPERTISE

cariousustaceo

View Profile

Stacey Waters

Corporate Security Officer

Dept: Computer Science and Engineering

Capacity: 0/8

EXPERTISE

pelisuppono

View Profile

Dr. Z

Associate Professor

Dept: CSE

Capacity: 3/5

EXPERTISE

Not specified

View Profile

Dr. Aina Rahman

Associate Professor

Dept: Computer Science and Engineering

Capacity: 4/8

Frank Holmes

Faculty Member

Dept: Computer Science and Engineering

Capacity: 0/0

Jesus Muhammad

Professor

Dept: Computer Science

Capacity: 0/0